



DEPARTMENT OF
COMPUTER SCIENCE

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Master in Computer Science and Engineering

COST AND OPERATIONS OPTIMIZATION IN MULTI-CLOUD ENVIRONMENTS OVER GENAI FRAMEWORKS

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NOVA University Lisbon

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ABSTRACT

Keywords: Multi-Cloud · Generative AI · Cost Optimisation · Cloud Operations · GenAI Frameworks

RESUMO

Palavras-chave: Multi-Cloud · Inteligência Artificial Generativa · Otimização de Custos · Operações em Nuvem · Frameworks GenAI

DISCLOSURE OF DELEGATION TO GENERATIVE ARTIFICIAL INTELLIGENCE

Duarte José Lourenço Belo declares¹ the use of Generative AI in the research work and writing process of this document. According to the GAIDeT taxonomy [33], the following tasks were delegated to Generative AI tools under full human supervision:

Conceptualization

- | | | |
|---|---|---|
| ☒ Idea generation | ☒ Formulating research questions and hypotheses | ☐ Feasibility assessment and risk evaluation |
| ☒ Defining the research objective | | ☐ Preliminary hypothesis testing |

Literature Review

- | | | |
|---|--|---|
| ☒ Literature search and systematization | ☒ Writing the literature review | ☐ Evaluation of novelty and identification of gaps |
| | ☐ Analysis of market trends and/or patent environment | |

Methodology

- | | | |
|---|---|--|
| ☒ Research design | ☒ Development of experimental or research protocols | ☐ Selection of research methods |
|---|---|--|

Software Development and Automation

- | | | |
|---|---|---|
| ☒ Code generation | ☐ Process automation | ☐ Creation of algorithms for data analysis |
| ☒ Code optimization | | |

Data Management

- | | | |
|---|---|--|
| ☒ Data collection | ☐ Data curation and organization | ☐ Reproducibility testing |
| ☒ Validation | ☐ Data analysis | |
| ☒ Data cleaning | ☐ Visualization | |

¹This declaration was generated using Lua $\text{\LaTeX}$ [34] and the $\text{\LaTeX}$ package `aidisclose` [23].

Writing and Editing

- | | | |
|--|--|--|
| ☒ Text generation | ☐ Formulation of conclusions | ☐ Translation |
| ☒ Proofreading and editing | ☐ Adapting and adjusting emotional tone | ☐ Reformatting |
| ☒ Summarizing text | | ☐ Press releases and outreach materials |

Ethics Review

- | | | |
|---|--|--|
| ☒ Bias analysis and discrimination assessment | ☒ Ethical risk analysis | ☐ Data confidentiality monitoring |
| | ☒ Monitoring compliance with ethical standards | |

Supervision

- | | | |
|---|--|--|
| ☐ Quality assessment | ☐ Identification of limitations | ☐ Publication support |
| ☐ Trend identification | ☐ Recommendations | |

Generative AI tools used: ChatGPT-5.2, Gemini 3 Pro, Claude 4.5 Opus, and Perplexity 12.12.25.

Additional comment #1: The AI was used primarily for refining the code in Section 3, but the algorithm logic was derived manually.

Additional comment #2: This is a second additional comment!

Additional comment: This is a third additional comment! It is starred and, thus, does not get a number. Also, this comment is longer and, hopefully, will span to the next line!

Additional comment #3: This is the third non-starred additional comment!

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ACRONYMS

GenAI generative artificial intelligence (*pp. 1–3*)

LLM large language model (*pp. 1, 3*)

MCP Model Context Protocol (*pp. 1–3*)

RAG retrieval-augmented generation (*p. 3*)

INTRODUCTION

1.1 Context & Motivation

As enterprise adoption of generative artificial intelligence (GenAI) transitions from exploratory pilots to sustained production use, organisations face a growing set of operational challenges that existing frameworks do not fully address. The rapid proliferation of large language model (LLM)-based capabilities has exposed critical gaps in how intelligent applications are built, integrated, and governed in production environments: fragmented tool connectivity, limited cost visibility, and insufficient orchestration flexibility [6].

Maestro is a framework for enterprise GenAI applications in multi-cloud environments, developed at Closer Consulting. It was conceived to address exactly these operational realities by providing a structured layer for deploying and managing GenAI workloads across heterogeneous cloud infrastructure. Since Maestro is designed to be implemented across different production environments, the framework must be both adaptable and robust: it should accommodate diverse model providers, integration patterns, and infrastructure configurations without requiring fundamental redesign at each deployment.

Despite this ambition, three concrete problems currently limit the framework's effectiveness in production. First, tool and service integration remains brittle. In the absence of a standardised connectivity protocol, each new data source or capability requires bespoke integration work, increasing development overhead and reducing portability across deployments [1]. The Model Context Protocol (MCP), introduced by Anthropic in 2024, offers a compelling solution to this problem by defining a universal interface through which LLM hosts can discover and invoke external tools and resources [1]. Integrating MCP into Maestro would enable a plug-and-play connectivity model that scales across production environments without duplicating integration effort.

Second, cost observability is severely constrained. In the current deployment context, LLM infrastructure costs are consolidated under shared cloud resources, with no segmentation by agent, workflow, or request type. This architectural reality makes it impossible to attribute expenditure to specific system components, which in turn prevents principled optimisation decisions. Without granular cost telemetry at the level of individual invocations, teams operating Maestro in production cannot identify inefficient patterns,

enforce budget boundaries, or demonstrate the cost-effectiveness of GenAI capabilities to stakeholders [39]. The absence of cost observability is not merely an inconvenience; it is a structural barrier to responsible scaling.

Third, multi-agent orchestration within Maestro lacks the composability required to support complex, interdependent workflows. As GenAI applications grow in scope, they increasingly require the coordination of specialised agents operating in parallel or in structured sequences. Current orchestration approaches do not provide sufficient primitives for expressing these dependencies reliably, which constrains the range of applications that can be built on top of the framework [38].

This dissertation addresses these three problems in a unified and operationally grounded manner. By extending Maestro with MCP-based integration, a cost observability layer, and an improved multi-agent orchestration model, this work contributes both a set of concrete framework improvements and a methodology for evaluating production-oriented GenAI infrastructure.

1.2 Institutional Context

The research presented in this dissertation emerges from a partnership between the Department of Computer Science at NOVA School of Science and Technology (FCT NOVA) and Closer Consulting, which serves as the main industrial collaborator. As the author, I have assumed a dual capacity—both as an academic researcher and as an active practitioner within Closer Consulting—enabling unique integration of theoretical inquiry and practical implementation.

Closer Consulting is responsible for the design, development, and maintenance of the Maestro GenAI framework, which forms the technical foundation and primary case study for this work. Throughout the research process, I have been granted privileged access to Maestro’s production deployments, real-world cost telemetry, and multi-tenant enterprise configurations, facilitating a level of analysis and validation grounded in operational reality [10].

This collaboration extends beyond resource access, shaping the research agenda and ensuring that outcomes directly address genuine industry needs. All architectural decisions, implementation steps, and evaluation metrics have been established within the constraints and requirements of authentic enterprise environments, rather than exclusively through controlled laboratory experiments.

1.3 Problem Statement

Despite the growing maturity of enterprise GenAI development, frameworks designed for production use continue to face a set of structural limitations that constrain their scalability, operational efficiency, and portability across deployment environments. This section identifies the three core challenges that motivate the present work, all of which

are grounded in the operational reality of Maestro, a framework for enterprise GenAI applications in multi-cloud environments developed at Closer Consulting [10].

The first challenge concerns the standardisation of tool and service integration. Modern enterprise GenAI systems are required to consume data from a wide variety of sources — databases, cloud services, analytics platforms, and enterprise APIs — across heterogeneous infrastructure. In Maestro’s current architecture, each such integration is implemented as a bespoke connector, with no shared interface contract governing how tools and resources are discovered or invoked by the underlying LLM [10]. This approach introduces significant friction: onboarding a new data source requires duplicated integration effort, integration logic is tightly coupled to the specific provider being consumed, and the resulting system is difficult to port across production environments without substantial rework. The MCP, introduced by Anthropic in 2024, proposes a universal, open protocol that standardises how LLM hosts discover and invoke external tools and resources [1, 25]. However, the integration of MCP into production-grade GenAI frameworks — including the definition of reusable server implementations across cloud platforms — remains an open engineering challenge with limited empirical treatment in the literature.

The second challenge concerns cost observability and operational efficiency. In the current deployment context of Maestro, LLM infrastructure costs are consolidated under shared cloud resources without segmentation by agent, workflow, or request type. This architectural reality means there is no mechanism to attribute expenditure to specific system components, which in turn prevents any principled decision-making about cost optimisation. Without granular cost telemetry at the level of individual invocations, it is not possible to identify inefficient request patterns, enforce budget boundaries per use case, or demonstrate the financial sustainability of GenAI capabilities to stakeholders [39]. Beyond the absence of observability, the literature identifies a broader set of cost reduction techniques — including prompt compression [16, 27], semantic caching [5], dynamic model routing [9], and retrieval-augmented generation (RAG) pipeline optimisation [14] — that have not been evaluated or implemented within the framework. The lack of both measurement infrastructure and applied optimisation represents a compounded barrier to responsible scaling in production [2].

The third challenge concerns orchestration composability for multi-agent workflows. As GenAI applications evolve from single-turn question-answering systems towards complex, multi-step reasoning pipelines, the need for robust multi-agent orchestration primitives becomes increasingly critical [38]. The current orchestration layer in Maestro consists of purpose-built, hand-rolled agents for specific tasks — sentiment analysis, session summarisation, relevance checking — implemented without a shared compositional model [10]. This design limits the range of workflows that can be expressed, makes it difficult to introduce new capabilities without invasive changes to the orchestration logic, and provides no standard mechanism for inter-agent communication, tool delegation, or controlled parallelism. As a consequence, the framework cannot readily support the class of complex, interdependent multi-agent applications that are increasingly demanded in

enterprise settings [38].

1.4 Objectives

Four research objectives are defined to address the structural limitations identified in Section 1.3. Each objective is stated as a concrete, measurable outcome with an explicit success criterion and a designated primary toolset. Where validation has not yet been completed, hedged language is used to signal that the claimed outcome is a research hypothesis subject to empirical confirmation.

O1 — MCP server layer for Maestro. The first objective is to design and implement an MCP-compliant server library that exposes Maestro’s microservice capabilities through the Model Context Protocol [1] [25]. The implementation will use the MCP specification [25] as primary tooling. By replacing the current $N \times M$ bespoke adapter structure with an $N + M$ protocol-mediated architecture, it is hypothesised that integration complexity will decrease substantially as the service catalogue grows. The objective will be considered fulfilled when a representative set of heterogeneous service integrations has been validated under the new protocol layer.

O2 — Unified cost telemetry pipeline. The second objective is to instrument Maestro’s production orchestration layer with a unified LLM and infrastructure cost telemetry pipeline, providing the first such integration for this platform. Langfuse will serve as the LLM observability layer, capturing per-request token consumption, per-model cost, and cache hit rate at the invocation level. Azure Monitor will aggregate infrastructure-level signals via OpenTelemetry, creating a unified operational view across both LLM and compute dimensions. The objective will be considered fulfilled when LLM invocation paths receive systematic instrumentation and a representative production cost baseline has been established. This baseline is a prerequisite for the qualitative evaluation of O3.

O3 — Cost reduction via tokenomics optimisation. The third objective is to evaluate and implement a set of cost-reduction techniques drawn from the three-layer tokenomics framework that structures the cost analysis of this thesis. Three complementary mechanisms will be applied: semantic caching via MongoDB Atlas Semantic Cache, cascade routing via LangChain model routing, and prompt compression via candidates like LLM-Lingua [16], TOON [35], and others. The target outcome is a measurable reduction in cost-per-query relative to the O2 baseline, with answer relevancy remaining qualitatively stable between the experimental and control conditions. The FrugalGPT cascade routing strategy [9] and GPTCache semantic caching [5] provide the conceptual grounding for the mechanisms evaluated here.

O4 — Composable agent graph orchestration. The fourth objective is to refactor Maestro’s orchestration layer using LangGraph as the primary orchestration runtime. LangGraph is selected for three reasons that are architecturally significant. First, Maestro already uses LangChain [8] in production; LangGraph is LangChain’s native graph orchestration runtime, making it an extension of the existing stack rather than a replacement, and

thereby minimising migration risk. Second, Azure AI Foundry natively hosts LangGraph agents, preserving alignment with the existing Azure infrastructure. Third, Langfuse integrates natively with LangGraph, meaning that O2 and O4 share the same instrumentation stack without additional tooling overhead — a deliberate architectural decision that eliminates the cost and complexity of maintaining parallel observability pipelines. The objective will be considered fulfilled when a representative set of multi-step production workflows has been refactored as composable LangGraph agent graphs, their execution steps receive systematic coverage by Langfuse traces, and the refactored implementations achieve a measurable reduction in code volume relative to their hand-crafted equivalents.

1.5 Expected Contribution

The work undertaken in pursuit of the four research objectives is expected to produce the following original contributions:

- An open, protocol-compliant MCP server library for Maestro, enabling standardised integration with cloud-native data services across Azure, AWS, and GCP and reusable by any MCP-compliant agent framework.
- A unified LLM and infrastructure cost telemetry pipeline — Langfuse combined with Azure Monitor via OpenTelemetry — instrumented across Maestro’s production orchestration layer, constituting the first such integration for this platform.
- Empirical benchmarks of cost-reduction techniques — semantic caching, LLM cascade routing, and prompt compression — evaluated on a real multi-cloud, multi-tenant production environment, providing ecologically valid estimates of technique efficacy that laboratory settings cannot offer.
- A reference architecture for composable, observable, and cost-aware enterprise GenAI frameworks in multi-cloud environments, documented as a reusable design pattern applicable beyond the Maestro platform.
- A quantitative LLM Ops baseline — comprising token cost, latency, and quality metrics for a production RAG system — establishing a reproducible evaluation methodology for future comparative studies in this domain.

1.6 Structure

BACKGROUND

2.1 Foundations of Neural Language Representation

The ability to represent language in a form that machines can process and reason over is a prerequisite for any modern Generative AI system. Two landmark contributions established the theoretical and practical foundations upon which contemporary large language models (LLMs) and retrieval-augmented architectures are built: the introduction of dense word representations via neural networks, and the development of the Transformer architecture based solely on attention mechanisms.

2.1.1 Distributed Word Representations

Prior to 2013, natural language processing relied predominantly on sparse, high-dimensional representations such as one-hot encodings, which offered no intrinsic notion of semantic similarity between words. Mikolov et al. [24] addressed this limitation by proposing Word2Vec, a computationally efficient framework for learning continuous vector representations of words from large unlabelled corpora. The model is trained through a self-supervised prediction task using one of two architectures: Continuous Bag-of-Words (CBOW), which predicts a target word from its surrounding context, or Skip-gram, which performs the inverse prediction.

The resulting word embeddings encode both syntactic and semantic structure in a dense vector space, enabling linear algebraic operations over meaning - most notably the well-known relationship $\vec{k}ing - \vec{m}an + \vec{w}oman \approx \vec{q}ueen$.

The significance of Word2Vec extends well beyond the paper itself. By demonstrating that semantic similarity can be captured as geometric proximity in a continuous vector space, it established the distributional hypothesis as an actionable engineering principle. This insight directly underlies the design of modern embedding models and vector databases, which are central components in Retrieval-Augmented Generation (RAG) pipelines [21]. In the context of this work, the quality and cost of the embedding stage in the Maestro framework’s RAG pipeline is a direct consequence of choices made in this representational paradigm.

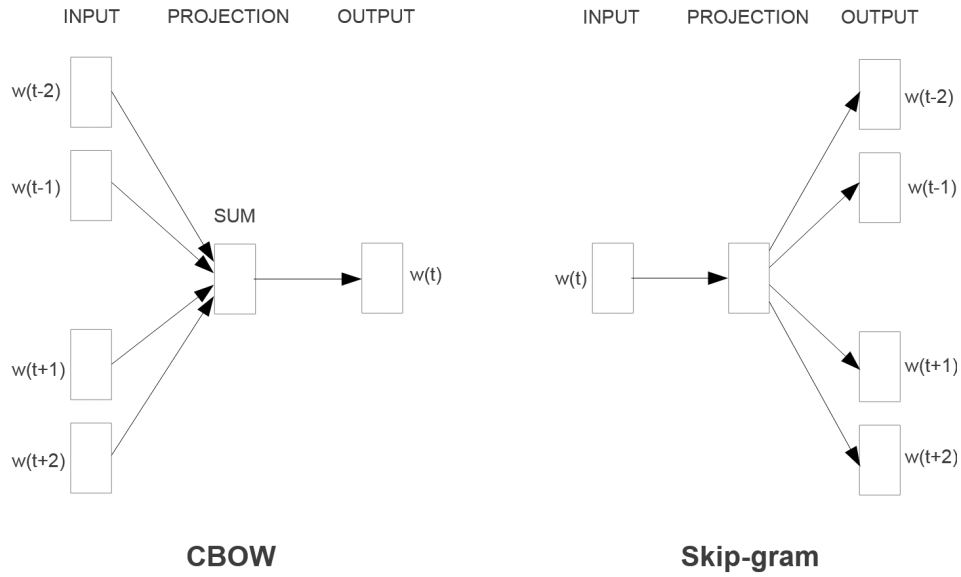


Figure 2.1: CBOW and Skip-gram architectures of the Word2Vec framework. Adapted from [24].

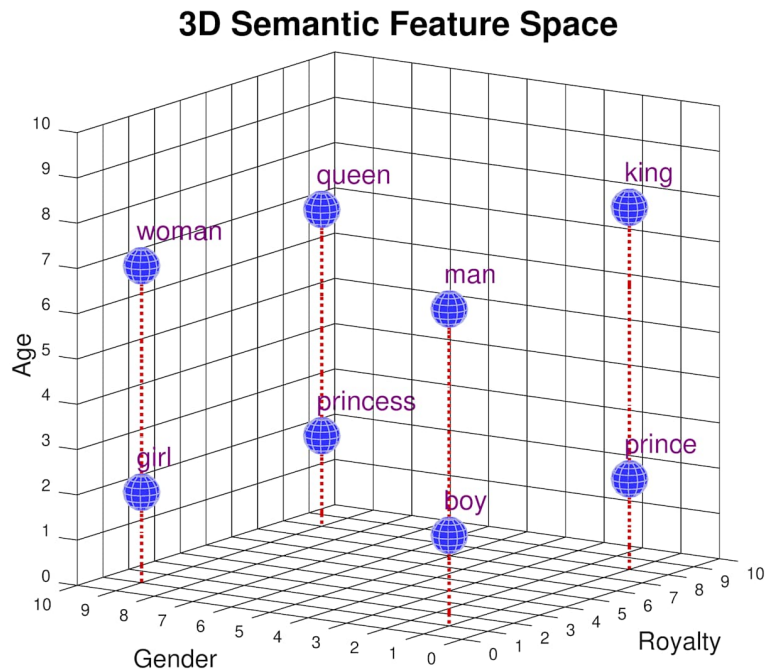


Figure 2.2: Word vector space learned by Word2Vec, illustrating semantic analogies such as $\vec{king} - \vec{man} + \vec{woman} \approx \vec{queen}$. Adapted from [24].

2.1.2 The Transformer Architecture

Despite the advances in distributed representations, sequence modelling remained constrained by recurrent architectures such as LSTMs and GRUs, which process tokens sequentially and suffer from vanishing gradients over long contexts. Vaswani et al. [37] proposed a radical departure: the Transformer, an architecture that dispenses with recurrence entirely and relies solely on attention mechanisms to model dependencies between all positions in a sequence simultaneously. The core operation, scaled dot-product attention, computes output representations as a weighted sum of value vectors, where weights are derived from the compatibility between query and key projections:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) V$$

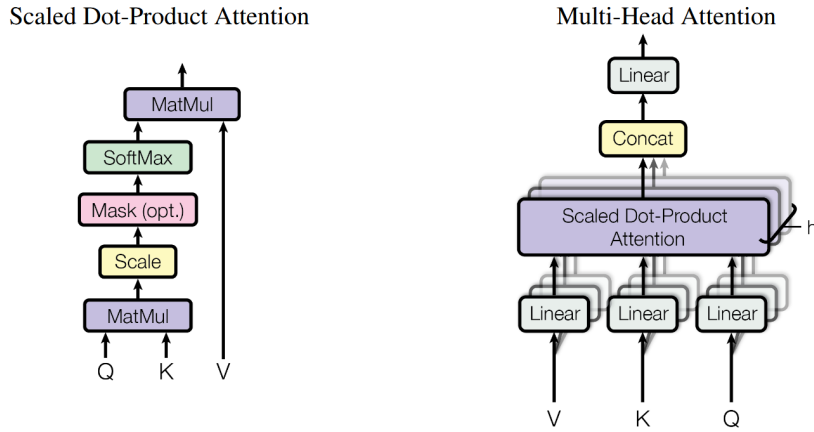


Figure 2.3: Scaled Dot-Product Attention (left) and Multi-Head Attention (right). Adapted from [37].

By running multiple such attention operations in parallel - Multi-Head Attention - the model learns to simultaneously attend to different types of relationships across the input, such as syntactic dependencies and semantic co-references. Positional encodings are injected additively into the input embeddings to compensate for the architecture's inherent permutation invariance. The resulting model achieved 28.4 BLEU on the WMT 2014 English-to-German translation benchmark, surpassing all prior approaches while requiring a fraction of the training compute [37].

The Transformer's parallelism over sequence positions unlocked a new scaling regime for neural language models. Architectures derived from it - most notably encoder-only models such as BERT [11] for representation learning, and decoder-only models such as the GPT series [6] for text generation - became the standard building blocks of enterprise GenAI systems. Critically, the attention mechanism scales quadratically with sequence length, $O(n^2)$, meaning that context length is a primary cost driver in token-priced LLM APIs. This architectural property is foundational to the FinOps strategies explored in this

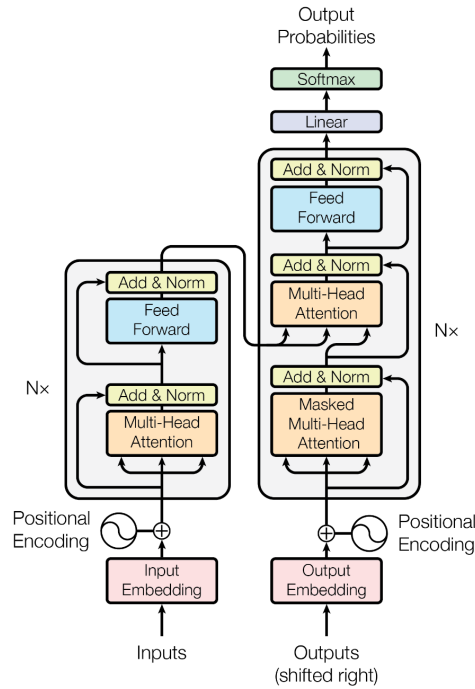


Figure 2.4: Full Transformer architecture with encoder and decoder stacks. Adapted from [37].

dissertation, namely prompt compression, semantic caching, and dynamic model selection, as each seeks to reduce the effective n processed by the model or the frequency with which inference is invoked at all [24, 37].

Together, these two works define the representational substrate of modern GenAI: Word2Vec established that meaning can be encoded as position in a learned vector space, and the Transformer established that context-sensitive reasoning over such representations can be performed efficiently and at scale. The remainder of this chapter examines the architectural patterns, operational frameworks, and cost optimisation techniques that build upon these foundations in production-grade enterprise systems.

2.2 Large Language Models in Production

The preceding section established the two foundational breakthroughs that made modern language AI possible: distributed word representations and the Transformer architecture. Building upon these primitives, the field witnessed a rapid consolidation around a unified paradigm - the large language model (LLM) - in which ever-larger Transformer-based networks, pre-trained on internet-scale corpora, acquire general-purpose language competence that can be redirected toward downstream tasks with minimal additional supervision. This section characterizes LLMs as a class, traces the methodological evolution from raw pre-training to instruction-following alignment, and examines the architectural and economic implications of deploying such systems at enterprise scale.

2.2.1 Large Language Models: An Overview

A large language model is a neural language model of sufficient parameter count and training data scale that general linguistic and reasoning capabilities emerge as a consequence of the pre-training objective alone, without task-specific architectural modification [6]. Pre-training proceeds via self-supervised next-token prediction over corpora of hundreds of billions to trillions of tokens, optimizing the model to assign high probability to naturally occurring text. The resulting parameter weights encode an implicit representation of syntax, semantics, world knowledge, and rudimentary reasoning chains that can be elicited through natural language prompting.

LLMs are predominantly realized as *decoder-only* Transformer architectures [37], in which the self-attention layers apply a causal mask that restricts each token’s attention to its leftward context, preserving the autoregressive generation property. This design choice contrasts with the encoder-only variant typified by BERT [11], which produces bidirectional contextual representations optimized for discriminative tasks. For generative applications - intelligent assistants, automated reasoning, and document synthesis - the decoder-only paradigm dominates, and it is this class of model that underpins the enterprise GenAI systems examined in this dissertation.

A defining empirical finding in the development of LLMs is the existence of *scaling laws*: systematic, power-law relationships between model size, training compute, dataset volume, and downstream performance [6]. Scaling laws imply that performance is largely predictable and improvable by increasing any of the three axes, establishing a framework within which investment in compute translates into measurable capability gains. This property has driven the successive scaling of commercial models and, critically, has also made token consumption a dominant cost variable: larger models process the same input more expensively, directly motivating the cost optimization strategies central to this work.

2.2.2 From Pre-training to Instruction Tuning

Despite the generative competence acquired during pre-training, early LLMs exhibited a systematic misalignment between model behavior and user intent. A model trained solely to predict the next token on a web-scale corpus learns to continue text in statistically plausible ways, but not necessarily in ways that are *helpful*, *honest*, or *harmless* when deployed as an interactive assistant. Ouyang et al. [26] addressed this gap by introducing a three-stage fine-tuning pipeline centered on Reinforcement Learning from Human Feedback (RLHF).

The procedure operates as follows. First, a *supervised fine-tuning* (SFT) phase trains the base model on high-quality demonstrations of desired behavior, produced by human labelers on a curated prompt distribution. Second, a *reward model* is trained to predict human preference scores over pairs of model outputs, internalizing a proxy for human judgment. Third, the SFT model is further optimized against the reward model signal using Proximal Policy Optimization (PPO), an online reinforcement learning algorithm. The resulting InstructGPT models, despite comprising as few as 1.3 billion parameters,

were consistently preferred by human evaluators over the unaligned 175-billion parameter GPT-3 baseline, demonstrating that alignment quality is not strictly a function of scale [26]. This result has profound practical implications: it established RLHF as the standard post-training procedure for production LLMs and confirmed that an aligned smaller model can match or exceed the utility of a larger unaligned one - a relationship that informs the dynamic model selection strategies explored in this dissertation.

2.2.3 Open-Weight Models and the Deployment Spectrum

The commercial LLM landscape is stratified between proprietary, API-accessed models and openly released, self-hostable alternatives. The former class, exemplified by OpenAI's GPT-4 family and Anthropic's Claude series, exposes inference through token-priced cloud APIs; the latter is represented most prominently by Meta AI's Llama series. Touvron et al. [36] introduced Llama 2, a family of models ranging from 7 to 70 billion parameters, pre-trained on two trillion tokens and released under a license permitting commercial use. Fine-tuned variants - Llama 2-Chat - further apply RLHF alignment, achieving performance competitive with closed-source alternatives on standard benchmarks while remaining locally deployable [36].

The existence of capable open-weight models introduces a consequential degree of freedom into enterprise GenAI architectures. Rather than committing uniformly to a single proprietary provider, organizations may route inference dynamically across a portfolio of models differentiated by capability tier, latency characteristics, and per-token cost. Smaller open-weight models, deployed on-premise or on reserved cloud instances, can serve high-volume, lower-complexity queries at substantially reduced marginal cost, while API-accessed frontier models handle tasks requiring peak reasoning performance. This model routing paradigm - often referred to as *LLM cascading* or *dynamic model selection* - is a central cost optimization mechanism evaluated in this dissertation in the context of the Maestro framework's multi-cloud deployment [36, 26].

Together, the progression from raw pre-training through RLHF alignment to the open-weight deployment spectrum defines the operational environment within which enterprise GenAI frameworks must function. The ability to select, combine, and switch between models of varying capability and cost is not merely an architectural convenience but a financial imperative in production systems where token consumption can scale three to five times per client without deliberate optimization [26].

2.3 Retrieval-Augmented Generation

Large language models store factual knowledge implicitly within their parameters, an approach that introduces well-documented limitations: parametric knowledge is frozen at training time, cannot be transparently audited, and tends to produce hallucinations when queries exceed the model's training distribution [21]. Retrieval-Augmented Generation

(RAG) addresses these limitations by coupling a generative model with a dynamic, non-parametric memory that can be queried, updated, and inspected at inference time. The following subsections trace the theoretical foundations of this paradigm, survey its subsequent architectural evolution, and ground the discussion in the dense retrieval machinery that makes scalable semantic search tractable in practice.

2.3.1 Foundational Architecture: Lewis et al. (2020)

The canonical RAG formulation was introduced by Lewis et al. [21], who proposed a general-purpose fine-tuning recipe for generative models endowed with access to an external non-parametric memory. The architecture decomposes into two principal components: (i) a *retriever* $p_\eta(z | x)$, implemented as a Dense Passage Retriever (DPR) [19] whose bi-encoder architecture maps both queries and documents into a shared dense vector space via independent BERT encoders, and (ii) a *generator* $p_\theta(y_i | x, z, y_{1:i-1})$, implemented as BART-large [20], which conditions its output sequence on the concatenation of the original query x and retrieved passages z .

Formally, the top- K passages are selected by solving a Maximum Inner Product Search (MIPS) problem over a pre-indexed Wikipedia corpus of 21 million 100-word chunks:

$$p_\eta(z | x) \propto \exp(d(z)^\top q(x)), \quad d(z) = \text{BERT}_d(z), \quad q(x) = \text{BERT}_q(x). \quad (2.1)$$

Two marginalization strategies over the latent document variable z are proposed. In *RAG-Sequence*, a single retrieved document is assumed responsible for the entire output sequence:

$$p_{\text{RAG-Seq}}(y | x) \approx \sum_{z \in \text{top-}k} p_\eta(z | x) \prod_i p_\theta(y_i | x, z, y_{1:i-1}). \quad (2.2)$$

In *RAG-Token*, a distinct document may be selected at each generation step, enabling the generator to synthesize content from multiple retrieved passages:

$$p_{\text{RAG-Token}}(y | x) \approx \prod_i \sum_{z \in \text{top-}k} p_\eta(z | x) p_\theta(y_i | x, z, y_{1:i-1}). \quad (2.3)$$

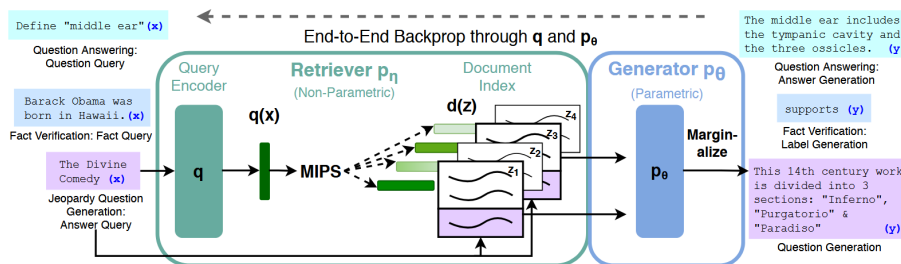


Figure 2.5: End-to-end RAG architecture combining a DPR retriever with a BART-large generator. Adapted from [21].

The retriever and generator are trained end-to-end via minimization of the negative marginal log-likelihood $\sum_j -\log p(y_j | x_j)$, without any direct supervision on which documents should be retrieved. Crucially, the document encoder and its FAISS index are kept frozen during fine-tuning; only the query encoder and the BART generator are updated, substantially reducing computational cost.

Empirically, RAG established state-of-the-art results on three open-domain question answering benchmarks-Natural Questions (44.5 EM), WebQuestions (45.2 EM), and CuratedTrec (52.2 EM)-outperforming both purely parametric seq2seq baselines (e.g., T5-11B) and task-specific retrieve-and-extract systems (e.g., DPR-based readers) [21]. For generative tasks, human evaluators rated RAG outputs as more factual than a BART baseline in 42.7% of Jeopardy question generation trials, compared to only 7.1% in which BART was judged superior. An index hot-swapping experiment further demonstrated that updating the non-parametric memory by replacing the document index is sufficient to update the model’s world knowledge without any retraining, a property of practical significance for production deployments.

2.3.2 Architectural Evolution: Naive, Advanced, and Modular RAG

Gao et al. [14] systematize the rapid evolution of RAG methods into a three-tier taxonomy, providing a structural lens through which the design of contemporary systems can be evaluated.

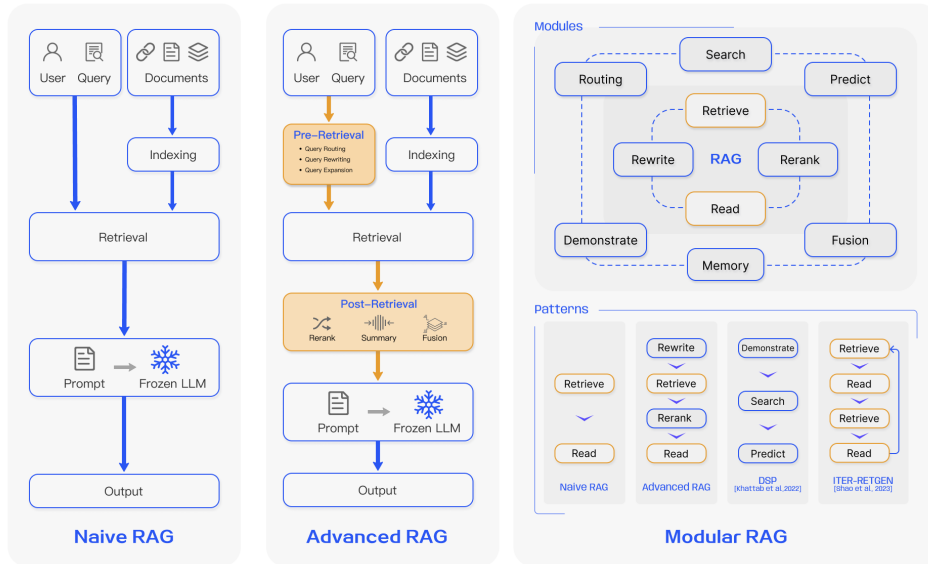


Figure 2.6: Three-tier RAG taxonomy: Naive, Advanced, and Modular RAG. Adapted from [14].

Naive RAG. The earliest post-ChatGPT RAG implementations follow a minimal *Retrieve-then-Read* pipeline consisting of three sequential steps: (1) *indexing*, in which raw documents are chunked, encoded via an embedding model, and stored in a vector database; (2) *retrieval*,

in which the top- K chunks most similar to the user query are identified via cosine similarity over dense embeddings; and (3) *generation*, in which the query and retrieved chunks are concatenated into a prompt supplied to the language model [14]. While effective in simple scenarios, Naive RAG exhibits well-characterized failure modes: low retrieval precision due to semantic gaps between queries and documents, context window saturation from uninformative passages, and hallucinations that arise when the generator defaults to parametric knowledge in the presence of noisy retrieved context.

Advanced RAG. Advanced RAG introduces targeted interventions at both the pre-retrieval and post-retrieval stages to address the precision and coherence limitations of the naive approach [14]. Pre-retrieval strategies include query rewriting, query expansion into sub-questions, and the construction of *Hypothetical Document Embeddings* (HyDE), in which the model first generates a hypothetical answer and then uses its embedding to retrieve semantically closer passages. Post-retrieval strategies include cross-encoder reranking to elevate the most relevant passages and context compression to eliminate redundant or off-topic content prior to generation. Indexing enhancements—such as sliding-window chunking, fine-grained segmentation, and metadata attachment—further improve retrieval recall and temporal awareness. These modifications maintain the linear Retrieve-then-Read structure of Naive RAG while substantially improving end-to-end generation quality.

Modular RAG. Modular RAG supersedes the fixed-pipeline paradigm by decomposing the system into independently composable functional modules: *Search*, *Memory*, *Routing*, *Predict*, and *Task Adapter* [14]. This architecture supports non-sequential execution patterns, including iterative retrieval loops (e.g., ITER-RETGEN [32]), adaptive retrieval triggered by generation uncertainty (e.g., FLARE [17]), and self-reflective retrieval with token-level reflection signals (e.g., Self-RAG [4]). The modular design also enables seamless integration with complementary techniques—fine-tuning the retriever, fine-tuning the generator, or training both collaboratively—without requiring architectural changes to the pipeline. Gao et al. [14] position Modular RAG as the current frontier, noting that its flexibility allows task-specific orchestration patterns that static pipelines cannot accommodate.

A practical implication of this taxonomy is that RAG and fine-tuning are not mutually exclusive. RAG excels in dynamic environments where knowledge must be updated frequently and interpretable provenance is required, whereas fine-tuning is better suited for scenarios demanding stylistic adaptation or the internalization of domain-specific reasoning patterns. In knowledge-intensive tasks, Gao et al. [14] report that RAG consistently outperforms unsupervised fine-tuning, both on knowledge encountered during pre-training and on entirely novel factual information.

2.3.3 Dense Semantic Retrieval: Sentence-BERT

The retrieval component of any RAG pipeline depends critically on the quality and computational tractability of sentence-level embeddings. Vanilla BERT representations are unsuitable for large-scale semantic search: because BERT processes sentence pairs jointly through a cross-attention encoder, identifying the most similar pair in a corpus of n sentences requires $O(n^2)$ inference passes, which Reimers and Gurevych [30] report as approximately 65 hours for $n = 10,000$ sentences. Furthermore, averaging BERT token embeddings or using the [CLS] token output yields representations that perform worse on semantic textual similarity (STS) tasks than simple GloVe embeddings, owing to BERT’s lack of explicit metric learning for sentence-level geometry.

Reimers and Gurevych [30] address these limitations by introducing *Sentence-BERT* (SBERT), a modification of BERT using Siamese and triplet network structures to produce fixed-size, semantically meaningful sentence embeddings amenable to efficient cosine-similarity comparison.

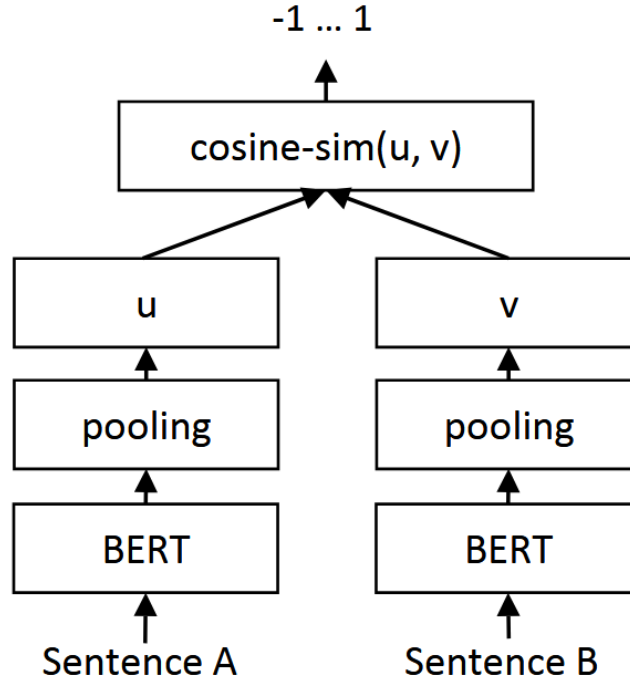


Figure 2.7: SBERT Siamese network architecture with triplet loss objective. Adapted from [30].

The architecture passes each sentence independently through a shared BERT encoder, then applies mean-pooling over the output token embeddings to obtain a single vector representation u and v for a sentence pair. Three training objectives are supported depending on the available supervision signal:

- **Classification objective** (for NLI corpora): a softmax classifier over the concatenation $[u; v; |u - v|]$ trained to predict entailment, contradiction, or neutral labels.
- **Regression objective** (for STS corpora): cosine-similarity between u and v optimized with mean-squared error against human-annotated similarity scores.

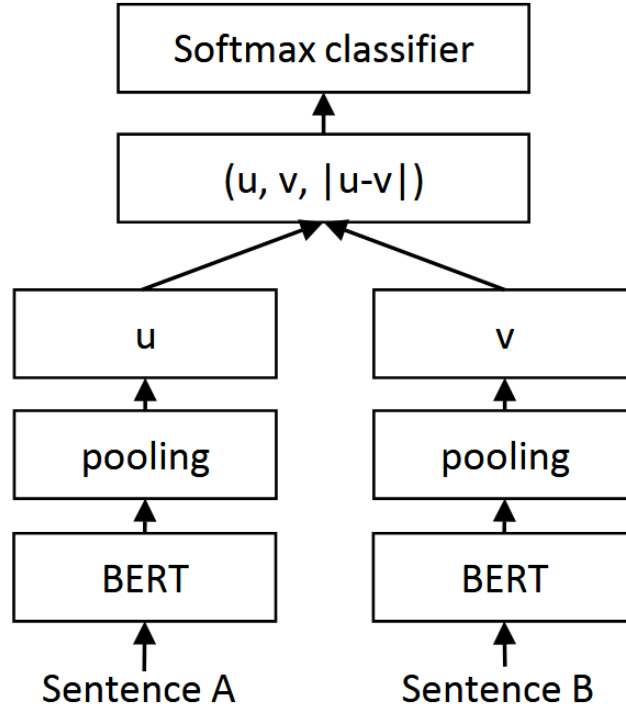


Figure 2.8: Classification and regression training objectives of SBERT. Adapted from [30].

- **Triplet objective**: a margin-based ranking loss that ensures the anchor embedding a is closer to a positive p than to a negative n by a margin ϵ :

$$\mathcal{L}_{\text{triplet}} = \max(\|a - p\| - \|a - n\| + \epsilon, 0). \quad (2.4)$$

The Siamese design decouples encoding from comparison: document embeddings are computed once at indexing time and stored in a vector index, reducing inference at query time to a single forward pass followed by a nearest-neighbor lookup. This reduces the 65-hour pairwise comparison to approximately 5 seconds for a 10,000-sentence corpus, while preserving-or improving upon-BERT’s accuracy on STS benchmarks [30]. On the STS tasks evaluated, SBERT achieves an improvement of 11 Spearman correlation points over InferSent and 5 points over Universal Sentence Encoder.

2.3.4 Connection to LlamaIndex

The theoretical contributions reviewed above map directly onto the abstractions provided by LlamaIndex, the framework employed in this work. LlamaIndex operationalizes the RAG pipeline of Lewis et al. [21] through a modular architecture that cleanly separates ingestion, indexing, retrieval, and synthesis [22]. During the *ingestion* stage, `SimpleDirectoryReader` loads raw documents, which are subsequently segmented by `SentenceSplitter` into overlapping chunks—a direct implementation of the chunking strategies discussed by Gao et al. [14] for Advanced RAG. The resulting nodes are encoded via a configurable embedding model (`Settings.embed_model`), which defaults to OpenAI’s `text-embedding-3-small` but is readily substituted with any SBERT-compatible model from the HuggingFace ecosystem (e.g., `BAAI/bge-small-en-v1.5`), whose architecture descends directly from the Siamese network paradigm of Reimers and Gurevych [30].

At query time, LlamaIndex’s `VectorStoreIndex` retrieves the top- K most relevant nodes via cosine similarity—the same MIPS formulation as DPR in [21]—and passes the retrieved context together with the user query to the configured language model for response synthesis. The framework further exposes advanced retrieval patterns consistent with the Modular RAG taxonomy of Gao et al. [14]: `RouterQueryEngine` implements semantic routing to multiple sub-indexes; post-retrieval reranking modules (e.g., `SentenceTransformerRerank`) apply cross-encoder scoring to improve precision; and iterative retrieval agents can be composed from LlamaIndex’s `ReActAgent` infrastructure. In this sense, LlamaIndex provides a production-grade instantiation of the entire RAG design space surveyed by Gao et al. [14], grounded in the dense retrieval foundations established by Lewis et al. [21] and Reimers and Gurevych [30].

2.4 Frameworks, MCP and Maestro

The foundations established in the preceding sections—dense vector representations, attention-based language models, retrieval-augmented generation, and dense semantic retrieval—describe the *components* of a modern GenAI system. What they do not describe is how those components are composed, governed, and operated at production scale. This section addresses that gap across three layers of increasing practical specificity.

2.4.1 LLM Orchestration Frameworks

The emergence of capable large language models created an immediate engineering challenge: how to compose LLMs with external tools, persistent memory, and retrieval systems into coherent, repeatable pipelines. LangChain [8] addressed this challenge by introducing a layered abstraction model built around three core primitives. A *chain* is a composable sequence of calls—to an LLM, a retriever, a tool, or another chain—that can be assembled declaratively and reused across applications. A *tool* is any callable capability exposed to the model with a name, a natural-language description, and a typed input schema;

the LLM uses these descriptions to decide, at runtime, which tool to invoke and with what arguments. *Memory* modules attach conversational context to any chain, enabling stateful interactions across multiple turns. Taken together, these primitives define what has become the dominant orchestration pattern in the field: a model-centric loop in which the LLM acts as a reasoning engine that selects and sequences external capabilities.

The theoretical foundation for this loop was established independently by [41] [41], who proposed the REACT paradigm—an acronym for *Reason + Act*. Rather than treating chain-of-thought reasoning and tool use as separate concerns, REACT interleaves them into a single iterative cycle: the model emits a *Thought* (an explicit reasoning step), selects an *Action* (a tool call), receives an *Observation* (the tool’s output), and repeats until a terminal condition is satisfied. Evaluated on multi-step knowledge-intensive tasks—including HotpotQA, FEVER, ALFWorld, and WebShop—REACT outperformed both chain-of-thought prompting alone and action-only baselines, demonstrating that the tight coupling of reasoning and acting is essential for robust agent behavior [41]. This loop is the intellectual ancestor of every agent executor in subsequent frameworks, including LangChain’s AgentExecutor, and it directly informs the orchestration design of the system examined in this work.

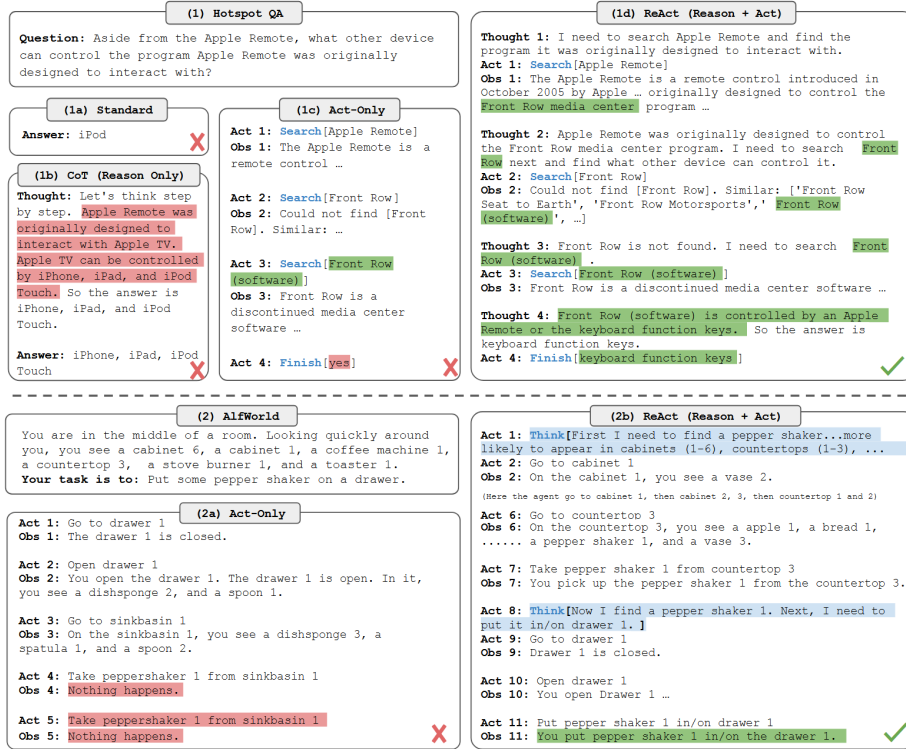


Figure 2.9: The REACT reasoning loop interleaving *Thought*, *Action*, and *Observation* steps. Adapted from [41].

2.4.2 Model Context Protocol

A persistent weakness of the orchestration paradigm described above is the absence of a standard interface between agents and the external systems they invoke. In practice, each integration is bespoke: a tool written for LangChain is not directly consumable by a different agent framework, and a function registered for OpenAI’s function-calling API is invisible to a Claude-based agent. This $N \times M$ integration problem— N agent frameworks multiplied by M external services—generates duplicated adapter code, inconsistent error handling, and fragile dependency chains that impede both development velocity and maintainability [1].

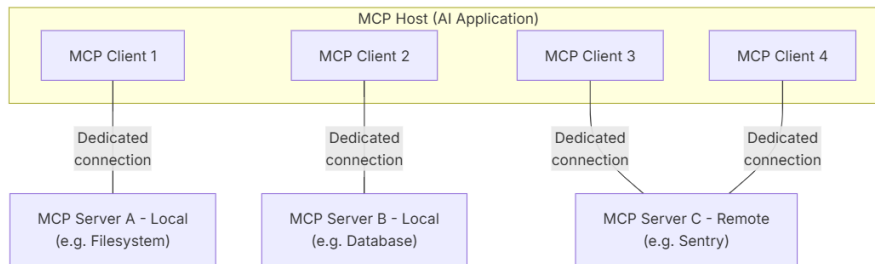


Figure 2.10: Model Context Protocol architecture showing the Host, Client, and Server roles connected via JSON-RPC 2.0. Adapted from [25].

The Model Context Protocol (MCP), introduced by Anthropic in 2024 [1] and specified in full at the canonical reference documentation [25], addresses this problem by defining an open, transport-agnostic protocol that standardizes how LLM-based hosts expose and consume external capabilities. The protocol organizes participants into three roles. An *MCP Host* is the application embedding the LLM—a development environment, a chat interface, or a custom agent runtime. An *MCP Client* is the component within the host that manages connections to one or more servers. An *MCP Server* is a lightweight process that exposes a bounded set of capabilities over JSON-RPC 2.0, using one of two transport mechanisms: standard I/O for local processes, or HTTP with Server-Sent Events for remote deployments [25]. Capabilities are organized into three first-class primitives: *Tools*—functions that the model may invoke with structured arguments, analogous to LangChain tools but protocol-defined; *Resources*—read-only data artifacts such as file contents, database schemas, or live API responses, exposed for model consumption without side-effects; and *Prompts*—parameterized prompt templates that servers can offer for reuse [1, 25].

The distinction between MCP and vendor-specific function calling is substantive. OpenAI’s function-calling mechanism, for example, is stateless, scoped to a single model provider, and requires the host application to manage all serialization, discovery, and transport concerns. MCP, by contrast, is stateful at the protocol level, provider-agnostic, and supports dynamic capability discovery: a client can query any MCP server for its current tool and resource catalog without prior knowledge of the server’s implementation. The consequence is a reduction from $N \times M$ to $N + M$ integration surface—any MCP-compliant

host can interoperate with any MCP-compliant server, regardless of the LLM or framework in use. This interoperability property is directly relevant to multi-cloud enterprise deployments, where the ability to expose heterogeneous data services-across Azure, AWS, and GCP-through a single protocol layer has significant implications for both governance and maintainability.

2.4.3 Enterprise GenAI Frameworks: Maestro in Context

Research prototypes built on frameworks such as LangChain and paradigms such as REACT operate under a set of assumptions that enterprise production environments do not share. In a production deployment serving multiple tenants across organizational units, correctness and capability are necessary but insufficient conditions. The system must additionally satisfy requirements for identity-based access control, secret management, encryption at rest and in transit, cost observability at the token and model level, multi-tenancy isolation, deployment portability across cloud providers, and operational reliability under variable load-none of which appear as first-class concerns in the orchestration literature reviewed above. Meeting these requirements demands an architectural discipline that goes beyond assembling chains and tools: it requires a governed platform in which each component is independently deployable, swappable without contract changes, and instrumented for both operational monitoring and audit compliance.

The Maestro GenAI Framework, developed by Closer Consulting, represents a practitioner response to precisely this gap [10]. Designed and validated across production deployments in the energy, healthcare, financial, and pharmaceutical sectors, Maestro is a multi-repository, container-native microservices platform whose architecture decomposes the RAG pipeline into independently governed services. The *LLM microservice* acts as a provider-agnostic gateway that routes generation requests to Azure OpenAI (GPT-4o, GPT-4o-mini), OpenAI, Azure AI Services (Mistral-Nemo, DeepSeek-R1), and Google Gemini, exposing a stable contract regardless of the underlying model. The *orchestrator* (chatbot-backend) implements the REACT-derived retrieval-augmented generation loop-history retrieval, hybrid vector search, per-chunk relevance filtering, prompt assembly, and LLM invocation-as a FastAPI service backed by LangChain. The *vector database service* encapsulates document ingestion, chunking, embedding via `text-embedding-ada-002`, and hybrid kNN+BM25 retrieval over MongoDB Atlas. The *datastore microservice* provides a provider-agnostic persistence layer over Azure Cosmos DB and MongoDB, covering chat history, feedback, and document metadata. A *guardrails microservice* implements PII detection using Microsoft Presidio for Portuguese-language inputs. Two *frontend surfaces*-a Streamlit chatbot with Microsoft Entra ID authentication and an Angular back-office for prompt and question management-complete the user-facing layer. Infrastructure is managed via OpenTofu targeting Azure, and observability is implemented through OpenTelemetry with Azure Monitor as the backend. In production, the framework has demonstrated measurable impact at scale: one deployment serving over 600 daily active users with

4,000 daily interactions across five embedded LLM models, and another achieving 83% question-answering accuracy over a corpus of 2,907 indexed legal documents [10].

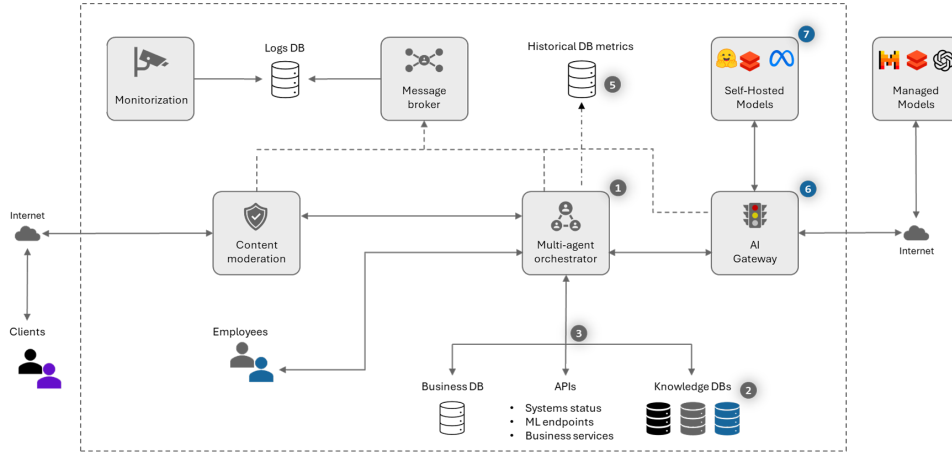


Figure 2.11: Functional architecture of the Maestro GenAI Framework, illustrating the *multi-agent orchestrator*, *AI Gateway*, *content moderation*, *self-hosted and managed model endpoints*, and *observability layer*. Adapted from [10].

In spite of this production maturity, a systematic analysis of the framework’s current implementation reveals a set of structural limitations that constrain its evolution along three dimensions: the absence of a standardized tool-exposure protocol—each microservice is reachable only through bespoke REST endpoints, with no dynamic capability discovery and no interoperability with external agent frameworks; an incomplete cost-observability and optimization layer—token usage and per-model cost are not exported as metrics, and no semantic caching or routing intelligence reduces redundant LLM invocations; and an agent orchestration architecture that relies on hand-crafted orchestrators rather than a composable, reusable agent graph. These three dimensions constitute the research frontier addressed in Section ??–?? of this work.

[TODO] Section range above uses placeholder labels; update references when those sections are written.

2.5 Multi-Cloud Computing and Infrastructure as Code

The deployment of production-grade GenAI systems at enterprise scale is not a purely algorithmic challenge. It is equally an infrastructural one. The components surveyed in the preceding sections - large language models, RAG pipelines, embedding models, and orchestration frameworks - must ultimately be provisioned, networked, secured, and operated on physical or virtual compute. The strategic choices made at this infrastructure layer directly influence system cost, resilience, and vendor dependency, each of which is a first-class design concern in the Maestro framework. This section examines the theoretical foundations of cloud computing and multi-cloud strategy, the interoperability and portability challenges that heterogeneous deployments introduce, the research landscape of

Infrastructure as Code as a discipline, and the role of declarative provisioning as the operational mechanism that makes reproducible, provider-agnostic infrastructure management tractable at scale.

2.5.1 Foundations of Cloud Computing

[3] identified three properties that fundamentally distinguish cloud computing from prior paradigms of enterprise infrastructure provision: the *illusion of infinite computing resources* available on demand, the *elimination of upfront capital commitments* by transferring hardware acquisition to the provider, and the ability to *pay for resources on a short-term basis* and release them when no longer required. These properties collectively enable a shift from capital expenditure (CapEx), in which organizations maintain underutilized on-premises hardware, to operational expenditure (OpEx), in which compute costs track actual consumption. Armbrust et al. empirically demonstrated that average server utilization in traditional enterprise data centers hovers between 5% and 20%, whereas elastic cloud provisioning allows organizations to closely track real demand and pay accordingly. This cost structure has direct implications for AI-intensive workloads, where inference load is characteristically bursty and difficult to predict in advance: the ability to provision and release compute resources on-demand without upfront commitment is precisely the property that makes multi-model, multi-provider GenAI deployments economically viable.

The same work identified a set of persistent obstacles to full cloud adoption, two of which are particularly relevant to this dissertation. *Data lock-in* arises when an organization’s data and application state become tightly coupled to a single provider’s proprietary storage and API formats, making migration prohibitively costly and creating asymmetric negotiating power in vendor relationships. *Service availability* refers to the concentration of operational risk in a single provider: outages, degradations, or unilateral policy changes propagate directly to all dependent workloads without recourse. Armbrust et al. identified the distribution of workloads across multiple independent providers as the principal mitigation for both failure modes [3]. This recommendation constitutes the foundational rationale for multi-cloud architectures and directly anticipates the design philosophy adopted in the Maestro framework, which provisions compute and model inference capacity across Azure, AWS, and GCP rather than committing uniformly to any single platform [10].

2.5.2 Interoperability and Portability in Multi-Cloud Environments

Deploying applications across heterogeneous cloud providers introduces a distinct class of engineering challenges that are not resolved by simply replicating resources across platforms. Pecteu [28] formalized these challenges by distinguishing two complementary but conceptually separate concerns. *Portability* denotes the capacity to transfer an application or workload from one cloud environment to another with bounded effort; *interoperability*

denotes the capacity for components deployed on different clouds to communicate and cooperate at runtime. Both properties require explicit architectural investment and neither emerges spontaneously from layering services across providers.

Petcu identified four primary sources of lock-in in multi-cloud deployments: proprietary APIs that expose non-standard resource abstractions; divergent data formats and serialization conventions that impede cross-provider data movement; incompatible service-level agreements that prevent unified governance policies; and asymmetric pricing models that obscure total cost of ownership in comparative analysis. Each source requires a distinct mitigation strategy — respectively, abstraction layers, protocol normalization, contractual harmonization, and unified billing instrumentation [28].

Petcu’s taxonomy further classifies multi-cloud deployment strategies along three operational archetypes. A *redundant* strategy designates one provider as active and one or more as standby, achieving resilience through replication at the cost of idle resource expenditure. A *complementary* strategy assigns distinct functional capabilities to distinct providers, exploiting each platform’s relative strengths without requiring full workload duplication. A *federated* strategy presents a unified abstraction to the client, with the underlying provider selection transparent to the consuming application [28]. Maestro operates primarily in the complementary mode: AWS Bedrock, Azure OpenAI, and Google Gemini are each exposed through a uniform llm-microservice interface that maintains a stable contract regardless of which backend processes a given inference request. This architecture directly instantiates the abstraction-layer pattern advocated by Petcu as the principal mechanism for resolving API heterogeneity across providers, while the stable service contract across providers is precisely what the interoperability axis of her framework prescribes [10].

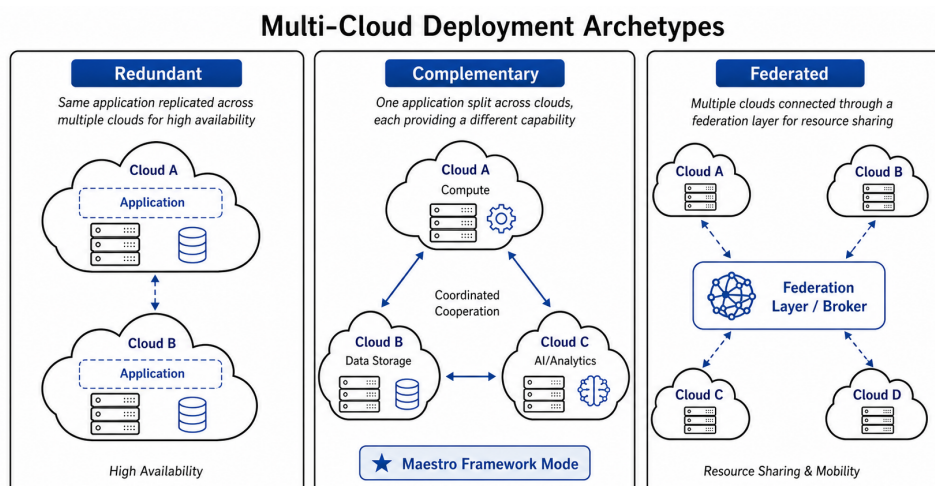


Figure 2.12: Multi-cloud deployment archetypes - redundant, complementary, and federated strategies - as classified by Petcu [28]. The Maestro framework operates in complementary mode.

[TODO] Replace the current AI-generated figure with an original diagram

2.5.3 Infrastructure as Code: Research Landscape and Practice

Infrastructure as Code (IaC) is the practice of expressing the configuration and provisioning of computing infrastructure - virtual machines, networks, storage, and managed services - as machine-readable, version-controlled source files interpreted by a provisioning engine, rather than as manual procedures executed by operators. As cloud adoption accelerated, IaC emerged as the primary mechanism for making infrastructure reproducible, auditable, and amenable to the engineering disciplines of code review and automated testing.

[29] conducted a systematic mapping study of the IaC research landscape, searching six scholarly databases and applying principled inclusion and exclusion criteria to identify 32 peer-reviewed publications directly concerned with IaC. Through qualitative analysis of this corpus, they identified four topics that collectively define the research agenda for the field: (i) *frameworks and tools for IaC*, encompassing the design and evaluation of provisioning engines and domain-specific languages; (ii) *use of IaC in practice*, examining adoption patterns and integration with broader DevOps workflows; (iii) *empirical studies of IaC*, including analyses of defect prevalence, security flaw introduction, and quality metrics in IaC scripts; and (iv) *testing in IaC*, addressing the challenge of validating infrastructure specifications prior to deployment [29]. A notable finding of the mapping study is that, at the time of publication, empirical research and testing remained substantially underexplored relative to tool-focused contributions - a gap that this dissertation partially addresses by empirically evaluating the cost and operational consequences of IaC-governed multi-cloud GenAI deployments.

Rahman et al. further observed that Ansible, Chef, Puppet, and Terraform were the dominant tools represented in the identified publications, and that research activity on IaC was growing rapidly in proportion to cloud adoption. Their characterization of IaC as a practice that “automatically configures system dependencies and provisions local and remote instances” captures the operational role that IaC plays in Maestro’s deployment architecture [29]: the framework’s infrastructure layer is managed via OpenTofu [10], the open-source, Linux Foundation-governed fork of HashiCorp Terraform, and thus falls squarely within the tool-focused and empirical categories that Rahman et al. identify as the core research vectors of the field.

OpenTofu operates on a declarative model in which practitioners specify the desired state of infrastructure resources - virtual networks, container registries, managed Kubernetes clusters, secret stores, and monitoring workspaces - and the engine computes a dependency-ordered execution plan to converge the actual state toward the declared target. The provider model maps directly onto the multi-cloud architecture described above: a single OpenTofu configuration references provider plugins for Azure, AWS, and GCP simultaneously, with each resource block scoped to the appropriate provider through namespace qualification. Because the provider abstraction insulates resource declarations from the underlying API differences documented by Petcu [28], operator-facing configuration remains uniform regardless of which cloud hosts a given resource - a property

essential for a framework that, by design, spans multiple providers.

Beyond provider abstraction, the declarative IaC approach enables two properties of direct relevance to this dissertation’s research objectives. First, it provides *environment reproducibility*: the same configuration that provisions a staging environment in one subscription can instantiate an identical topology in a production subscription or a second cloud region, eliminating configuration drift as a source of environment-specific defects. Second, it establishes a *cost-attribution surface*: because every provisioned resource is declared explicitly in versioned code, cost anomalies can be traced to specific configuration changes through commit history, supporting the FinOps disciplines of budget forecasting and spend attribution that are central to the optimisation objectives of this work [3, 10].

2.6 Cost Optimization

The economic profile of large language model (LLM) systems is dominated by inference costs that scale with the number of tokens processed, the capability tier of the selected model, and the frequency with which inference is invoked. In multi-cloud GenAI deployments, these costs accumulate across providers and workloads, making cost optimization a first-class architectural concern rather than a post-hoc operational adjustment [3, 28, 10]. This section surveys five complementary classes of techniques—semantic caching, model cascades, prompt compression, reranking, and prompt caching—together with the broader market context that shapes their impact.

At a high level, all of these techniques can be viewed as mechanisms for reducing the *effective token cost* of a system. Some reduce the number of tokens that reach a model (prompt compression, reranking), others reduce the number of model calls (semantic caching, prompt caching), and others reduce the marginal cost per call by routing traffic to cheaper models when possible (LLM cascades). In a multi-cloud setting, these mechanisms compose with provider-specific pricing to determine the realized cost of serving a given workload [14, 10].

A unifying lens for understanding all of these mechanisms is *tokenomics*: the discipline of minimising token consumption relative to the semantic value delivered by each inference call. The token is the atomic economic primitive of LLM-based systems—every capability, from factual recall to multi-step reasoning, is ultimately paid for in tokens whose price is governed by scaling-law dynamics [6]. Because the cost-accuracy trade-off is not fixed but can be shaped by architectural choice, as FrugalGPT demonstrates by achieving GPT-4-level accuracy at a fraction of its cost through adaptive routing [9], tokenomics is not merely a pricing concern but a systems-design discipline. Empirical evidence further shows that market forces alone are insufficient to resolve this problem: token prices have fallen substantially over time [2], yet total inference spend continues to grow as prompt lengths and usage volumes expand. Every technique surveyed in this section is therefore best understood not as an isolated optimisation but as a composable lever over a shared cost function, operating at one of three distinct layers: the *model layer* (cascade routing and

dynamic model selection), the *retrieval layer* (reranking and caching), and the *representation layer* (prompt compression, output style, and data serialisation).

2.6.1 Semantic Caching

Semantic caching is motivated by the observation that many production queries are not unique: users frequently repeat questions, rephrase earlier requests, or ask follow-up questions whose answers can be derived from previous responses. Traditional caches such as Redis operate at the level of exact key matching and cannot exploit semantic similarity, so even minor paraphrases lead to cache misses. GPTCache addresses this limitation by introducing an open-source *semantic* cache for LLM applications that stores responses and retrieves them via embedding-based similarity search instead of exact-string matching [5].

The architecture of GPTCache is conceptually straightforward. Incoming queries are first sent to GPTCache, which encodes both the query and previously stored queries and responses into vector embeddings using an embedding model compatible with vector databases such as Milvus or other ANN indexes [5]. A nearest-neighbour search over these embeddings determines whether a semantically similar query has already been answered; if the similarity exceeds a configurable threshold, the cached answer is returned directly. Only on a cache miss does the system forward the request to the underlying LLM, and the new query–response pair is then inserted into the cache for future reuse [5].

Empirical results reported in the GPTCache paper show that, when integrated with OpenAI’s GPT services, cache hits can increase response speed by a factor of 2–10, while also insulating the application from network fluctuations and provider-side latency spikes [5]. Because the cost of an embedding lookup is negligible compared to a full LLM call, a modest hit rate already translates into tangible API savings in token-priced environments. This is particularly relevant for enterprise assistants, FAQ bots, and RAG-based systems where user queries exhibit strong repetition patterns across tenants and time [5, 10].

For the multi-cloud framework studied in this dissertation, semantic caching aligns naturally with existing RAG infrastructure. The system already computes embeddings for retrieval and indexing; extending this embedding layer to back a semantic cache therefore requires minimal architectural change. In practice, semantic caching can be deployed as a cross-cutting concern in the LLM gateway, ensuring that repeated queries are intercepted before they reach any particular provider, thereby reducing both cost and latency across clouds [10, 22].

2.6.2 LLM Cascades

Where semantic caching reduces redundant *calls*, LLM cascades target the cost of *which* model is invoked when a call is necessary. FrugalGPT formalizes this idea by studying how to route queries across a portfolio of models to minimize cost subject to accuracy constraints. The central premise is that not all requests require the most capable (and

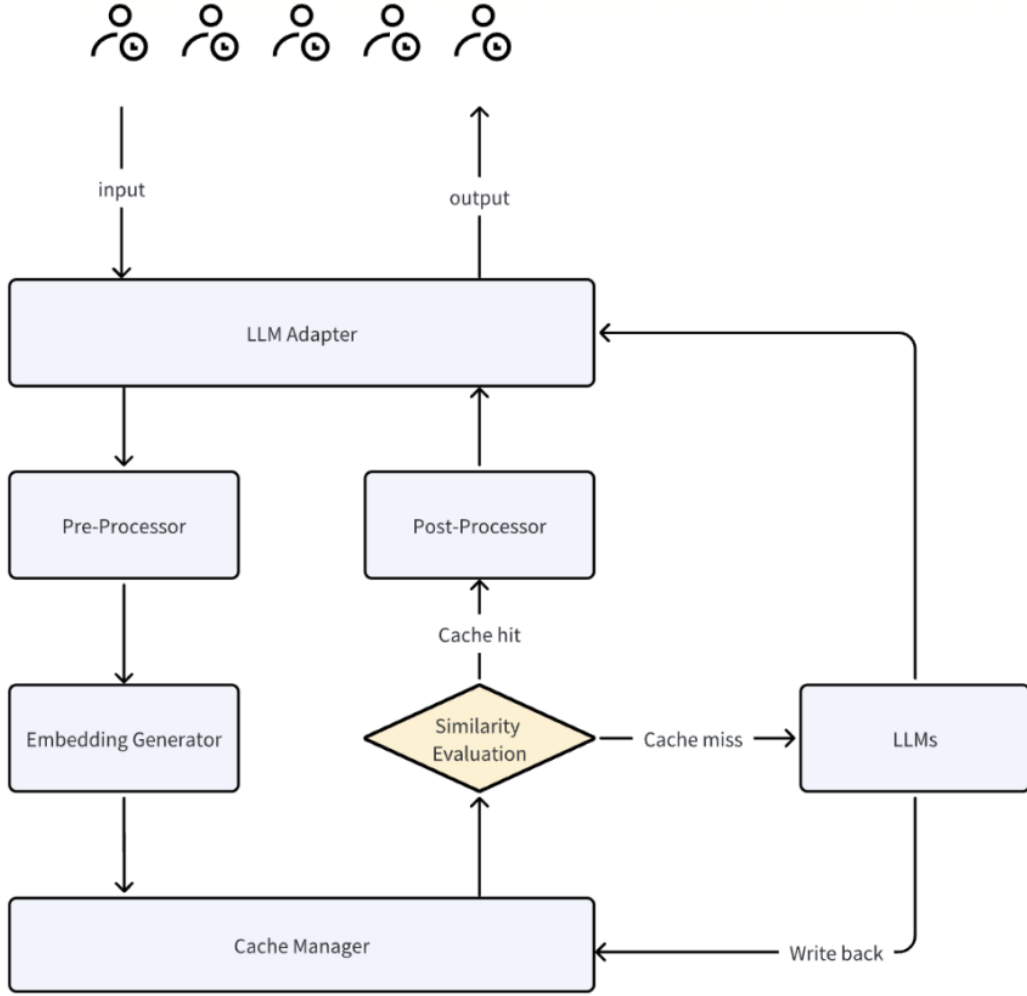


Figure 2.13: Architecture of GPTCache as a semantic caching layer for LLM applications, showing the adapter, pre-processor, embedding generator, cache manager, similarity evaluator, and post-processor. Adapted from Bang et al. [5].

expensive) model: many can be handled adequately by smaller, cheaper models, as long as the system can detect which queries are “easy” and which are “hard” [9].

FrugalGPT proposes several routing strategies, including rule-based routing based on query features, learned routing policies that predict which model will answer a query correctly, and multi-step cascades where a query is first answered by a cheap model and escalated only if the answer is judged unsatisfactory [9]. The experimental evaluation shows that such cascades can reduce inference cost by 50–98% while maintaining accuracy comparable to the best single model; in some configurations, FrugalGPT even exceeds the accuracy of GPT-4 while operating at the same or lower cost, by judiciously combining the strengths of multiple models [9].

This work is directly connected to the broader literature on model scaling laws and alignment. Scaling-law studies suggest that larger models generally perform better, but

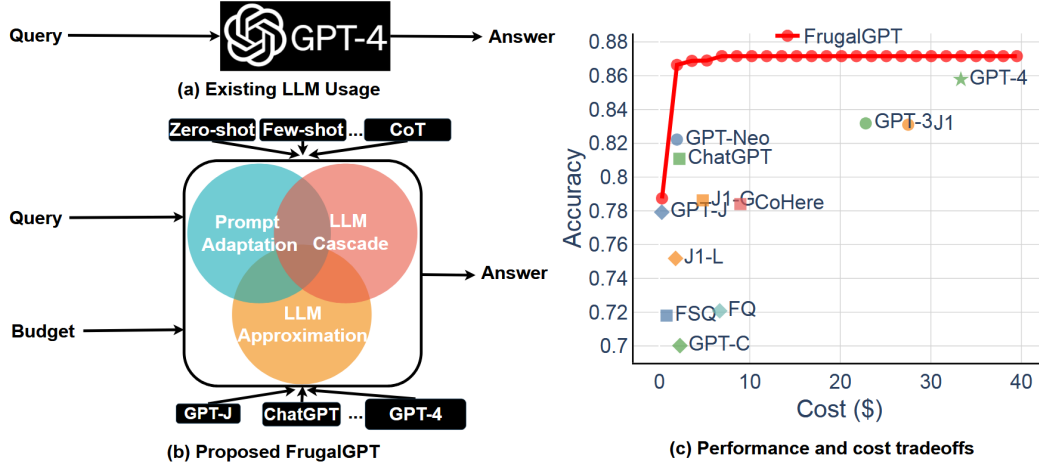


Figure 2.14: Overview of cost-reduction strategies in FrugalGPT, including prompt adaptation, LLM approximation, and LLM cascade. Adapted from Chen et al. [9].

at higher per-token cost [6]. At the same time, alignment research such as InstructGPT shows that smaller, well-aligned models can outperform larger unaligned ones on human preference metrics [26]. FrugalGPT leverages both observations by constructing cascades in which small, cheap models handle the bulk of traffic, while larger models are reserved for edge cases that truly benefit from their additional capacity [9, 26].

In a multi-cloud GenAI framework, cascades can be realized at the level of the LLM gateway. The gateway can expose a single logical endpoint to clients while internally routing requests among on-premise open-weight models (e.g., Llama 2), mid-tier cloud APIs, and frontier proprietary models, according to a routing policy informed by query characteristics, historical performance, and business priorities [36, 10]. This arrangement transforms model choice from a static configuration into a dynamic optimization problem over latency, accuracy, and cost, which can be tuned per tenant or per workload.

2.6.3 Prompt Compression

Prompt compression addresses a different cost driver: the number of tokens that must be processed per inference. Transformer-based LLMs have an attention mechanism whose computational and memory complexity scales quadratically with sequence length, meaning that longer prompts both cost more and are slower to process [37, 6]. In realistic enterprise workflows, prompt length is often dominated not by the user’s immediate query, but by system instructions, conversation history, and retrieved context from RAG pipelines [14, 10].

Research on prompt compression explores how much of this input can be removed or condensed without materially degrading model performance. Techniques range from heuristic truncation and summarization to learned compression models that identify the most salient segments of the input. Although the LLM-in-a-Flash paper by Jin et al. focuses

primarily on hardware-level optimizations—storing model weights on flash memory and using techniques such as windowing and row–column bundling to reduce data transfer—it also illustrates a broader point: that many aspects of LLM inference cost are driven by unnecessary data movement and redundant processing, both at the hardware and token levels [18, 37].

The LLM-in-a-Flash framework constructs an inference cost model that explicitly accounts for the characteristics of flash memory, then introduces two techniques to optimize inference under limited DRAM. First, windowing reuses previously activated neurons to reduce the amount of data that must be read from flash, and second, row–column bundling increases the contiguity of reads to better exploit the sequential-access strengths of flash [18]. Together, these techniques enable running models up to twice the size of available DRAM, with reported 4–5× speed-ups on CPUs and 20–25× speed-ups on GPUs compared to naive loading strategies [18].

For the purposes of this dissertation, the key analogy is that token-level prompt compression plays a similar role at the software layer as windowing and bundling do at the hardware layer: both reduce the amount of information the system must handle per request. In practice, prompt compression can be integrated in the orchestration framework by inserting a compression step between retrieval and generation, where the retrieved passages and conversation history are summarized or filtered before being passed to the LLM. This can significantly reduce token usage without sacrificing answer quality, especially in knowledge-intensive tasks with verbose supporting documents [14, 10].

At the software layer, the most academically grounded approach to prompt compression is the LLMingua family of methods. LLMingua, proposed by Jiang et al. at EMNLP 2023, uses a small surrogate model—such as GPT-2 or LLaMA-7B—to score the saliency of each token in the input via conditional perplexity [16]. Tokens that are highly predictable given their context contribute little unique information and can therefore be removed with minimal impact on the downstream task. The compression pipeline operates in three stages: a budget controller that determines the global compression ratio, an iterative token-level pruning step that removes low-saliency tokens subject to the budget, and an instruction-tuning alignment stage that recovers any quality degradation introduced by compression. Empirically, LLMingua achieves up to 20× compression while preserving the factual content required for the LLM to answer correctly [16]. LLMingua-2, proposed by Pan et al. (2024), reformulates compression as a token classification problem [27]. Rather than relying on unidirectional perplexity scoring, it trains a bidirectional encoder using data distilled from GPT-4, enabling the model to make retention decisions based on both left and right context. This formulation is task-agnostic, computationally faster at inference time, and produces compressions that are more faithful to the original meaning than the perplexity-based approach [27]. Both methods exploit the same fundamental gap in natural language: a large fraction of the tokens in any well-formed prompt are consumed by grammatical predictability—articles, auxiliary verbs, hedging phrases—rather than by the conveyance of factual content. Eliminating this predictable overhead is, in tokenomics

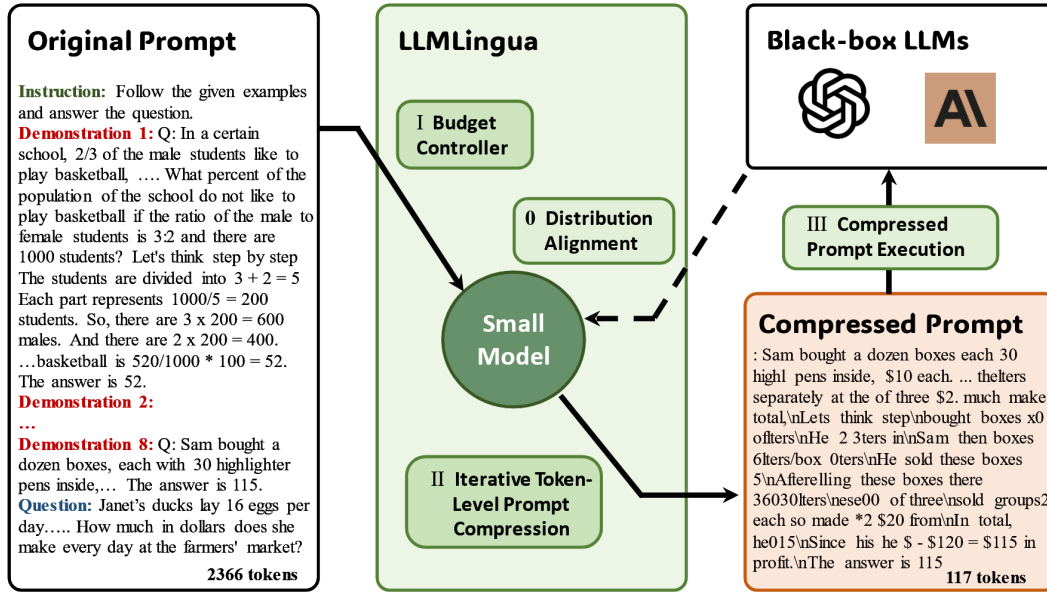


Figure 2.15: The LLMingua coarse-to-fine prompt compression framework, comprising a budget controller, iterative token-level compression, and a distribution alignment stage. The small surrogate model scores token saliency via conditional perplexity; low-saliency tokens are iteratively removed before the compressed prompt is forwarded to the target LLM. Adapted from Jiang et al. [16].

terms, equivalent to removing tokens whose marginal semantic value is near zero, a principle that will be revisited at the representation layer in Section 2.6.7.

2.6.4 Reranking Efficiency

Reranking techniques address the efficiency and quality of retrieval, which indirectly affects cost by influencing how much and what kind of context is included in the prompt. In naive RAG pipelines, the top- k documents from a vector search are simply concatenated with the user query and fed to the LLM. This approach is simple but brittle: if k is too small, relevant information may be omitted; if k is too large, the prompt becomes long and noisy, increasing both cost and the risk of hallucinations [21, 14].

Advanced RAG architectures introduce a reranking stage to mitigate these issues. For example, Gao et al. describe pipelines where an initial dense retriever proposes a candidate set, and a more expensive cross-encoder reranker then re-scores these candidates to select a smaller, higher-precision subset for inclusion in the prompt [14]. This two-stage design shifts some cost from the generator to the retriever: instead of asking the LLM to parse and ignore irrelevant passages inside a long prompt, the system spends a controlled amount of compute upfront to filter them out. When measured in effective token cost—the total number of tokens processed along the pipeline—reranking can therefore be cheaper than relying solely on inference-time reasoning over large contexts.

In the context of this work, reranking can be implemented as a dedicated microservice in the RAG subsystem. For instance, the vector store can return a larger candidate set (e.g., $k = 50$), which is then reranked using a cross-encoder model such as a Sentence-BERT variant fine-tuned for passage relevance, before selecting a smaller number of chunks (e.g., $m = 5\text{--}10$) to include in the prompt [30, 22]. This design improves the quality of retrieved evidence while keeping the prompt size under control, lowering both the monetary and computational cost of generation [14, 10].

2.6.5 Prompt Caching

Prompt caching is a complementary mechanism that has recently been introduced at the API layer by major model providers. The core idea is that repeated prompt prefixes—such as long system prompts, policy text, or shared instructions—can be cached on the provider side and billed at a discounted rate when reused across multiple requests. Anthropic’s Model Context Protocol (MCP) ecosystem and related documentation describe prompt caching as part of a broader effort to standardize how hosts and tools expose context and capabilities to LLMs, including the ability to reuse stable prompt templates efficiently [1, 25].

From the application developer’s perspective, prompt caching differs from semantic caching in two important respects. First, it operates at the level of *exact* prefix reuse rather than semantic similarity: the cached segment must be identical for the provider to apply the discount. Second, it does not bypass inference; instead, it makes inference cheaper on the repeated portion of the prompt while still processing the new suffix at full price. This makes it particularly attractive for enterprise assistants where a substantial portion of each prompt consists of fixed instructions or organization-specific policies that change rarely compared to user queries [1, 25].

In a multi-cloud architecture, prompt-caching support will vary across providers, but the design principles remain the same. The orchestration layer should be structured so that stable prompt components are explicitly separated from dynamic ones, and so that repeated prefixes can be reused across requests and sessions. Combined with semantic caching and cascades, prompt caching allows the system to benefit from both vendor-side and application-side reuse, jointly reducing latency and cost without compromising functional behavior [10, 8].

2.6.6 Market Context and LLMflation

The effectiveness and urgency of cost-optimization strategies are shaped by the broader dynamics of LLM pricing. Sardana and Frankle introduce the notion of “LLMflation” to describe the rapid decline in LLM inference costs over time, arguing that token prices have dropped by several orders of magnitude for comparable capability tiers as infrastructure, hardware, and model efficiency have improved [2]. Analyses of token price indices report

similar trends, documenting substantial reductions in the cost per million tokens across both proprietary and open-weight ecosystems.

However, lower unit prices do not automatically translate into lower total spend. As LLMs become cheaper and more capable, organizations tend to expand their usage, apply models to more workflows, and increase the amount of context supplied to each query, especially in RAG and agentic settings [14, 10]. In many production systems, demand growth and prompt length inflation can easily outpace price reductions, leading to absolute cost growth despite lower per-token rates. This dynamic closely mirrors historical patterns in cloud computing, where decreasing resource prices have often coincided with increasing overall expenditure due to workload expansion [3, 28].

For these reasons, the literature consistently argues that cost optimization must be embedded into the architecture of GenAI systems rather than treated as a transient concern that will be “solved” by future price cuts. Techniques such as semantic caching, model cascades, prompt compression, reranking, and prompt caching provide orthogonal levers that remain valuable even as prices fall, because they address structural drivers of cost: repeated queries, over-provisioned model capacity, unnecessary tokens, and inefficient retrieval [5, 9, 14, 2]. In a multi-cloud environment, these levers also interact with placement and portability decisions, which determine how workloads are distributed across providers and pricing regimes [28, 10].

2.6.7 Tokenomics in Practice: Representation and Style as Cost Levers

Beyond model-layer and retrieval-layer optimisations, token consumption is shaped by choices that precede or follow the model call itself: the format in which structured data is serialised into the prompt and the verbosity register of the output requested. These representation-layer decisions are less studied in the academic literature than caching or routing, but empirical evidence suggests they can contribute savings comparable in magnitude to software-layer compression techniques.

One practitioner-originating contribution at the representation layer is TOON (Token-Oriented Object Notation), proposed as a compact drop-in replacement for JSON in LLM prompts [35]. The mechanism exploits a structural inefficiency in JSON: for tabular payloads, every row repeats the full set of key names and syntactic delimiters, the majority of which are consumed by schema rather than data. TOON eliminates this redundancy by declaring column headers once and encoding rows as tab-delimited values, a format that retains the relational structure of the data while removing the per-row key verbosity. Practitioner benchmarks report token reductions of 30–60% on flat tabular structures without loss of semantic content [35]. This is particularly consequential for enterprise RAG systems that inject structured business data—product catalogues, API schemas, entity metadata—into prompts as JSON objects, where the overhead of key repetition can constitute a significant fraction of total prompt length.

A complementary representation-layer finding, originating in practitioner benchmarks

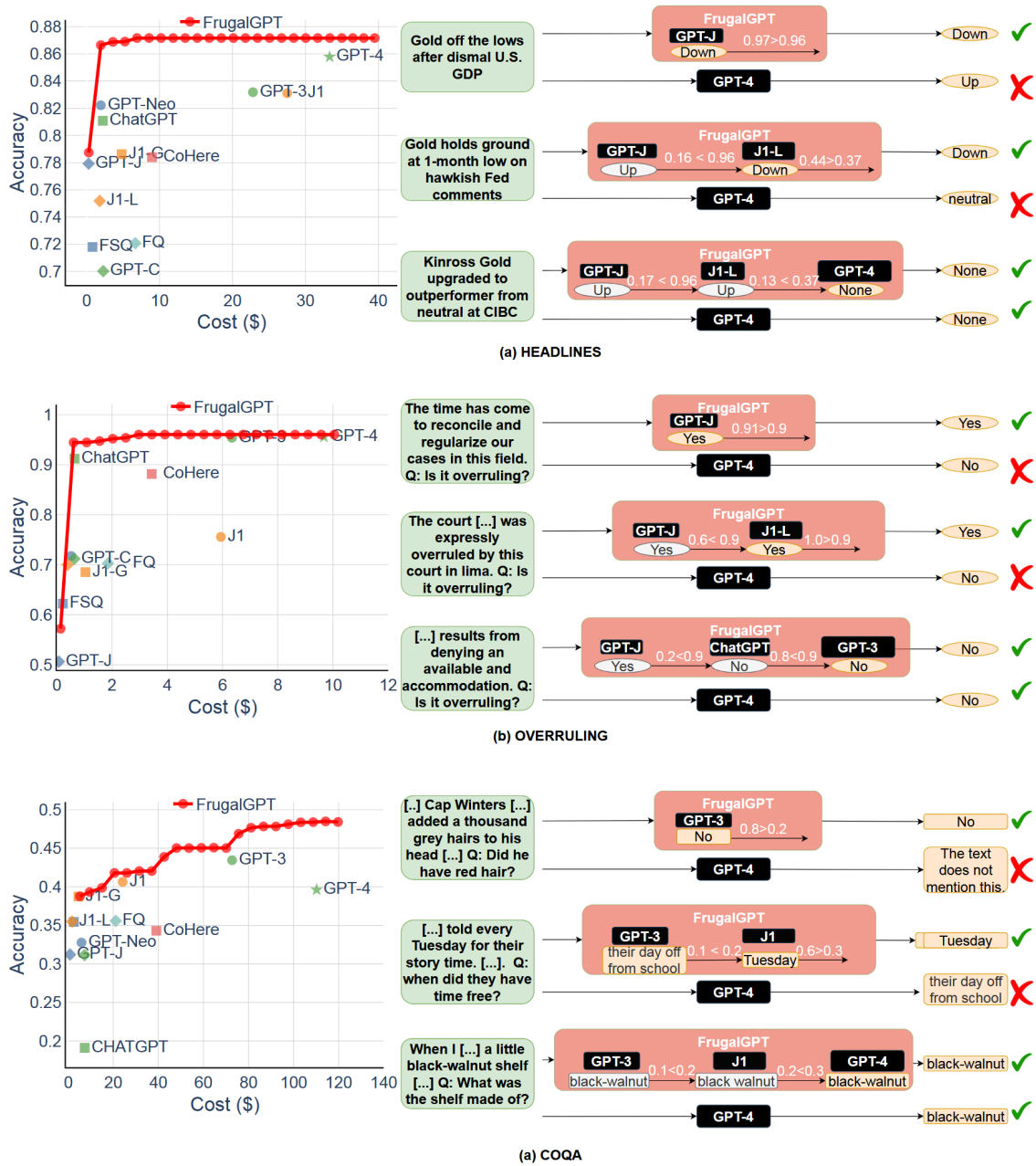


Figure 2.16: Accuracy and cost tradeoff achieved by FrugalGPT, showing that adaptive LLM cascades can reduce inference cost while maintaining or improving task performance. Adapted from Chen et al. [9].

rather than peer-reviewed literature, is the Caveman Compression technique [7]. The technique is predicated on the empirical observation that LLMs can produce factually equivalent responses in a telegraphic register that omits articles, auxiliary verbs, and hedging language, in the manner of a telegraphic message or early-stage note, without degrading factual accuracy or actionability. The original skill, a Claude Code prompt engineering contribution by Brussee (2025), achieved approximately 75% output token reduction on structured tasks [7]. A subsequent distillation by Guzik (2026) reduced

the meta-instruction itself from 552 to 85 tokens while still preserving 14–21% output token savings over a simple “Be concise” baseline, demonstrating that the technique is robust to significant compression of the instruction itself [15]. The mechanistic basis of Caveman Compression is consistent with the LLMlingua framing reviewed in Section 2.6.3: both techniques exploit the gap between the token budget consumed by grammatically predictable surface forms and the token budget strictly necessary to convey the factual payload. The distinction is directional—LLMLingua removes predictable tokens from the *input*; Caveman Compression removes them from the *output*—but both target the same underlying redundancy in natural language. It should be noted that these findings originate in practitioner benchmarks rather than peer-reviewed studies; they are included here as empirical evidence of the representation-layer tokenomics effect and as motivation for formal investigation.

The three-layer tokenomics model developed across this section can now be stated explicitly. At the *model layer*, cascade routing and dynamic model selection—as formalised by FrugalGPT [9]—reduce cost by ensuring that expensive frontier models are invoked only when their additional capacity is warranted. At the *retrieval layer*, semantic caching [5], prompt caching, and reranking reduce cost by preventing redundant inference calls and by filtering irrelevant context before it reaches the generator. At the *representation layer*, software-layer compression methods such as LLMlingua [16], format optimisations such as TOON [35], and output style control via techniques such as Caveman Compression [7] reduce cost by eliminating tokens whose marginal semantic contribution is low relative to their billing weight. This taxonomy is not merely descriptive: it frames the engineering contributions of this dissertation, in which the Maestro framework improvements are designed to address cost levers at all three layers within a unified multi-cloud architecture.

2.7 GenAI Operations and Governance

The deployment of Generative AI systems into production environments introduces a class of operational challenges that extends well beyond the concerns addressed by traditional software engineering. While earlier sections of this chapter have examined the architectural components of Retrieval-Augmented Generation pipelines and the cost dynamics of large language model inference, this section addresses the orthogonal question of how such systems are governed, monitored, and sustained over time. The central argument advanced here is that operational governance is not an optional layer applied atop a working system; it is a structural requirement whose absence compounds into systemic risk. This perspective is grounded in a foundational body of literature that has progressively refined the vocabulary and methodology for managing machine learning systems in production.

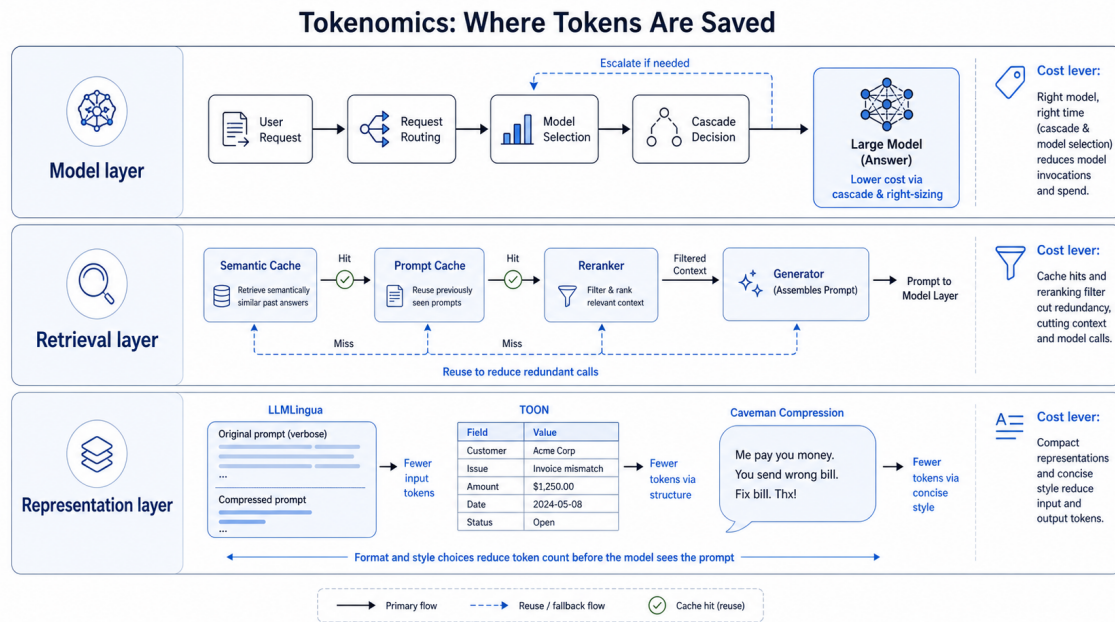


Figure 2.17: Three-layer tokenomics model for cost optimization in enterprise GenAI systems. The model layer controls cascade routing and dynamic model selection; the retrieval layer governs semantic caching, prompt caching, and reranking; the representation layer encompasses software compression (LLMLingua), format optimisation (TOON), and output style control (Caveman). Each layer targets a structurally distinct source of token consumption.

2.7.1 The Technical Debt Perspective on ML Operations

The conceptual foundation for understanding the governance challenges inherent in deployed machine learning systems was established by [31], whose seminal work catalogued the forms of *hidden technical debt* that accumulate in ML pipelines. The authors argue that ML code itself typically constitutes a small fraction of the overall system surface area; the dominant cost lies in the surrounding infrastructure: data ingestion pipelines, feature engineering routines, serving layers, monitoring components, and configuration management schemes. Crucially, this infrastructure is subject to a phenomenon the authors term *boundary erosion*, whereby the coupling between data distributions, model behaviours, and downstream consumers degrades silently over time, without triggering explicit failures.

Of particular relevance to RAG-based systems is the principle of CACE - "Change Anything, Changes Everything" - which describes the entangled nature of ML pipelines where altering a single upstream artefact, such as an embedding model or a document chunking strategy, may propagate unexpected behavioural changes through the entire system [31]. In the context of a production RAG framework such as Maestro, this entanglement is structurally amplified: the system integrates a continuously evolving knowledge base, a swappable retrieval layer, and a provider-agnostic LLM interface, each of which constitutes a potential source of silent degradation. The work of [31] thus provides the normative

justification for the governance objectives pursued in this thesis: without systematic instrumentation, evaluation, and operational discipline, the accumulation of technical debt in GenAI systems is not merely possible but inevitable.

2.7.2 Evaluating RAG Pipeline Quality in Production

A governance framework for RAG systems is only as rigorous as the evaluation methodology it employs. Classical generation metrics such as BLEU and ROUGE are inadequate for assessing the correctness of retrieval-augmented outputs, as they evaluate surface-level lexical overlap rather than factual faithfulness or contextual relevance [12]. To address this limitation, [12] introduced RAGAS (*Retrieval Augmented Generation Assessment*), a reference-free evaluation framework designed to assess RAG pipelines along multiple independent quality dimensions without requiring manually annotated ground truth.

The RAGAS framework decomposes pipeline quality into four core metrics: *faithfulness*, which measures the degree to which generated answers are grounded in the retrieved context; *answer relevance*, which assesses whether the response addresses the posed question; *context precision*, which evaluates whether the retrieved passages are focused on the relevant information; and *context recall*, which captures whether the retrieval covers the information necessary to produce a correct answer [12]. These metrics are computed using an LLM-as-judge paradigm, enabling automated and scalable evaluation that can be embedded directly into continuous integration and monitoring pipelines. The RAGAS framework serves as the methodological basis for the output quality assessment objective - designated O3 in this thesis - and provides the quantitative instrumentation through which Maestro's retrieval and generation quality is tracked in production.

2.7.3 RAGOps: Operationalising the Dual Lifecycle

The emergence of RAG as the dominant paradigm for enterprise LLM deployment has surfaced a class of operational challenges that existing LLMOps practices are insufficient to address. Xu et al. [40] introduce the concept of *RAGOps* to characterise the operational discipline required to manage RAG pipelines in production, arguing that such systems possess an inherently dual lifecycle: an *LLM lifecycle*, governing model versioning, prompt management, and inference configuration; and a *data lifecycle*, governing the evolution, quality, and freshness of the external knowledge base that the retrieval component depends upon.

This dual lifecycle framing is consequential for system design. In a conventional LLM deployment, the model artefact is the primary unit of management; its behaviour is stable between updates. In a RAG system, however, the external knowledge base undergoes continuous change - documents are ingested, revised, or deprecated; embedding distributions may shift as the indexing strategy evolves; and retrieval relevance may degrade independently of any model change [40]. Xu et al. [40] further characterise the architectural concerns of RAGOps using a 4+1 model view, providing a structured vocabulary

for reasoning about the logical, process, development, and physical dimensions of RAG system operations. Applied to the Maestro framework, this perspective motivates the need for automated mechanisms addressing knowledge base staleness, embedding index versioning, and retrieval quality monitoring - operational capabilities that are not routinely addressed by LLMOps tooling designed for static model deployments. Xu et al. [40] additionally report that approximately 60% of enterprise LLM deployments incorporate some form of retrieval augmentation, underscoring the industrial urgency of the RAGOps discipline.

2.7.4 Financial Governance: The FinOps Lifecycle for AI

The operational governance of GenAI systems necessarily encompasses an economic dimension. Token-priced inference introduces a cost model that is fundamentally more dynamic and usage-dependent than the resource-reservation models typical of classical cloud workloads. Without deliberate financial governance, LLM-based systems are susceptible to cost overruns driven by prompt inefficiency, suboptimal model routing, or unconstrained retrieval pipelines. The FinOps Foundation has extended its established cloud financial management framework to address these AI-specific dynamics, defining a three-phase governance cycle applicable to AI workloads: *Inform*, *Optimize*, and *Operate* [13].

In the *Inform* phase, organisations establish visibility into AI expenditure by instrumenting token usage, model call frequency, and cost per query. The *Optimize* phase leverages this visibility to identify and implement improvements, including model downsizing for lower-complexity queries, semantic caching, prompt compression, and retrieval efficiency enhancements. The *Operate* phase embeds these improvements into ongoing operational practice, establishing cross-functional accountability between engineering, product, and finance teams and closing the governance loop iteratively [13]. This cycle is continuous by design: the FinOps for AI framework explicitly positions cost governance not as a one-time optimisation exercise but as a persistent operational discipline that must be integrated into the deployment and monitoring pipeline. This framing directly informs the cost efficiency objective - designated O2 in this thesis - and provides the governance structure within which the cost reduction techniques examined in earlier sections are situated as components of a managed, accountable operational process rather than as isolated engineering interventions.

2.7.5 Synthesis: Towards a Governed GenAI Operations Model

The four perspectives reviewed in this section converge on a unified operational thesis: deploying a GenAI system is substantially less demanding than governing it sustainably in production. Sculley et al. [31] established the structural sources of operational risk in ML systems; Xu et al. [40] extended this analysis to the dual-lifecycle complexity specific to RAG architectures; Es et al. [12] provided the evaluation instrumentation necessary

to operationalise quality governance; and the FinOps Foundation supplied the financial governance framework within which quality and cost objectives are managed iteratively and accountably.

Taken together, these contributions define a layered governance model for production RAG systems, progressing from *structural risk awareness* (Sculley et al.) through *architectural discipline* (RAGOps) to *quality measurement* (RAGAS) and *financial accountability* (FinOps). The Maestro framework is positioned within this governance model as an instantiation of its principles in an enterprise multi-cloud context. The operational objectives O2 and O3 defined for this thesis - cost efficiency and output quality assurance, respectively - correspond directly to the Optimize and Inform–Operate dimensions of this model, grounding the empirical contributions of this work in an established theoretical and industry-validated governance architecture.

PROPOSED APPROACH

- 3.1 Overview**
- 3.2 Architecture**
- 3.3 Components and Design Decisions**
- 3.4 Cost and Operations Model**
- 3.5 Summary**

IMPLEMENTATION

4.1 Overview

4.2 Technology Stack

4.3 Framework Implementation

4.3.1 Component A

4.3.2 Component B

4.4 Deployment and Configuration

4.5 Summary

EVALUATION

5.1 Evaluation Methodology

5.2 Experimental Setup

5.3 Results

5.4 Discussion

5.5 Summary

CONCLUSION

6.1 Summary of Contributions

6.2 Limitations

6.3 Future Work

6.4 Final Remarks

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APPENDIX TITLE

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